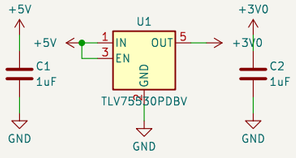
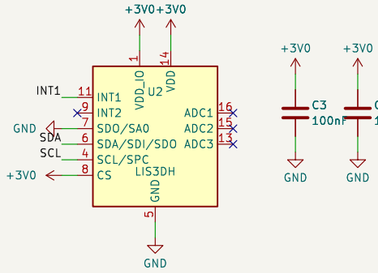


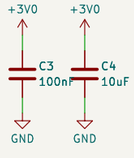
J1 -> NanoPC-T4 40-pin
1: VIN -> pin 2 (5V)
2: GND -> pin 6
3: SCL -> pin 5 (I2C2_SCL, 3V)
4: SDA -> pin 3 (I2C2_SDA, 3V)
5: INT1 -> pin 11 (GPIO1_A1, 3V, optional)
6: CHG_INT -> pin 7 (GPIO1_A0, 3V)
7: RFID_TX -> pin 21 (UART4_RX, 3V)
Do NOT use pins 27/28 (I2C2 1.8V)



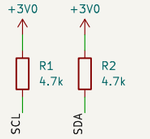
VIN 3.3-5.5V, 3.0V out
(matches T4 I2C2 3.0V I/O)



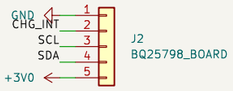
LIS3DH, I2C addr 0x18 (SA0=GND)
CS=VDD_IO selects I2C
C3/C4 close to pin 14



J3 -> WL-125 RFID (1x10 socket, 2.54mm)
J3 pin = module pin + 1
1: key (next to module pin 1), NC
2,3: ANT1/2 -> J4 antenna connector, ~100Vpp
4: ASCII select = VCC_RF (UART mode)
5: BEEP (NC)
6: TX 9600 8N1, 5V -> Q1 level shifter -> RFID_TX 3.0V
7: Wiegand D0 (NC)
8: /RESET (R7 pull-up)
9: GND
10: VCC_RF (5V, 3-stage RC filtered)

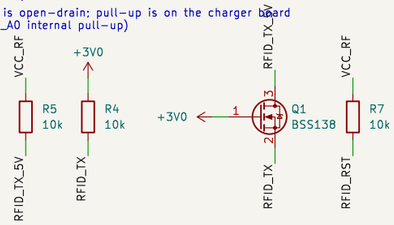


I2C pull-ups to 3.0V
(DNP if T4 pull-ups suffice)

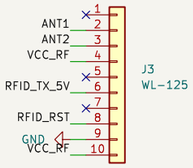


J2 -> BQ25798 charger board (JST PH 2.0)
1: GND
2: INT (BQ25798 /INT, open-drain, active low)
3: I2C_SCL
4: I2C_SDA
5: 3V (out, from U1)
BQ25798 I2C addr 0x6B

CHG_INT: BQ25798 /INT is open-drain; pull-up is on the charger board
(or enable the T4 GPIO1_A0 internal pull-up)

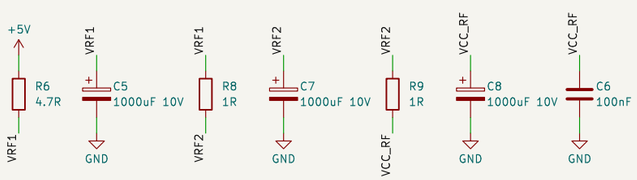


Q1/R4/R5: BSS138 level shifter (Adafruit #757 topology)
LV 3.0V (T4 UART4_RX) <-> HV 5V (WL-125 TX)





J4 -> RFID antenna coil (Molex PicoBlade 1.25mm, 2-pin)
ANT1/ANT2 carry ~50-100Vpp at 125kHz:
short, parallel traces; >=0.5mm clearance; no copper pour around them



WL-125 supply: 3-stage RC filter +5V -> VRF1 -> VRF2 -> VCC_RF
R6 4.7R limits inrush (~1A); drop ~0.17V @ 25mA
C8 + C6 close to J3 pin 10 (module VCC)



H1/H2: M3 mounting holes (3.2mm, NPTH)
mount near U2 so the accelerometer is held rigidly

Host: NanoPC-T4 I2C2 (3.0V I/O)

Sheet: /
File: lis3dh_board.kicad_sch

Title: LIS3DH I2C Accelerometer Board

Size: A3	Date: 2026-10-16	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1